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MODULE *TransCommit*

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CONSTANT *RM*

VARIABLE *rmState*

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$TCTypeOK \triangleq rmState \in [RM \rightarrow \{\text{"working"}, \text{"prepared"}, \text{"committed"}, \text{"aborted"}\}]$

$TCInit \triangleq rmState = [r \in RM \mapsto \text{"working"}]$

$canCommit \triangleq \forall r \in RM : rmState[r] \in \{\text{"prepared"}, \text{"committed"}\}$

$notCommitted \triangleq \forall r \in RM : rmState[r] \neq \text{"committed"}$

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$Prepare(r) \triangleq \begin{aligned} &\wedge rmState[r] = \text{"working"} \\ &\wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{"prepared"}] \end{aligned}$

$Decide(r) \triangleq \begin{aligned} &\vee \wedge rmState[r] = \text{"prepared"} \\ &\quad \wedge canCommit \\ &\quad \wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{"committed"}] \\ &\vee \wedge rmState[r] \in \{\text{"working"}, \text{"prepared"}\} \\ &\quad \wedge notCommitted \\ &\quad \wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{"aborted"}] \end{aligned}$

$TCNext \triangleq \exists r \in RM : Prepare(r) \vee Decide(r)$

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$TCConsistent \triangleq$

$\forall r1, r2 \in RM : \neg \wedge rmState[r1] = \text{"aborted"}$   
 $\quad \wedge rmState[r2] = \text{"committed"}$

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$TCSpec \triangleq TCInit \wedge \Box[TCNext]_{rmState}$

THEOREM  $TCSpec \Rightarrow \Box(TCTypeOK \wedge TCConsistent)$

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\\* Modification History  
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